



α -CBM : Label-Free Concept Bottleneck Models for Audio

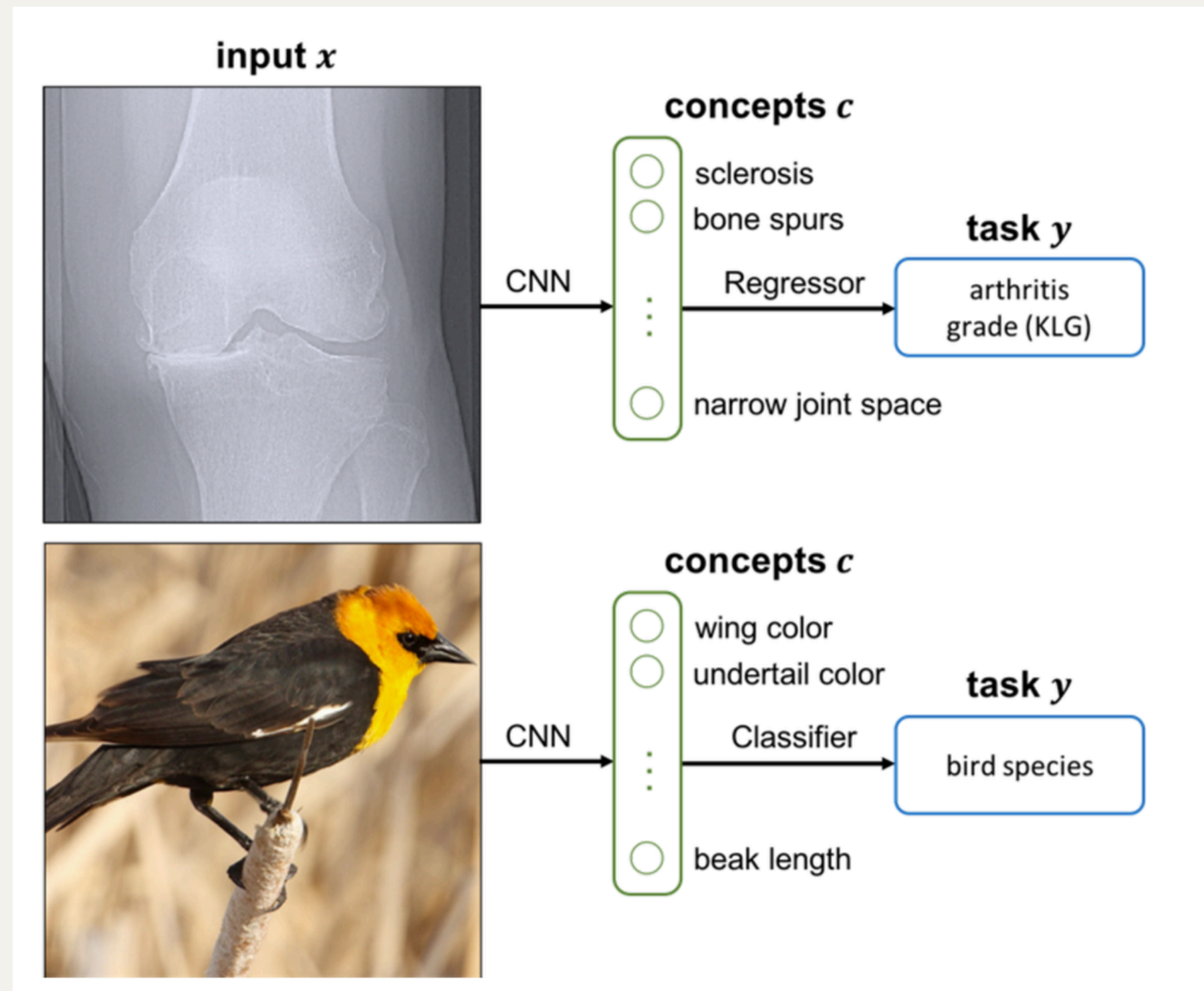
Amine Maazizi^{1,2}, Adam Gassem^{1,2}

¹MVA, ENS Paris-Saclay

²ENSTA Paris

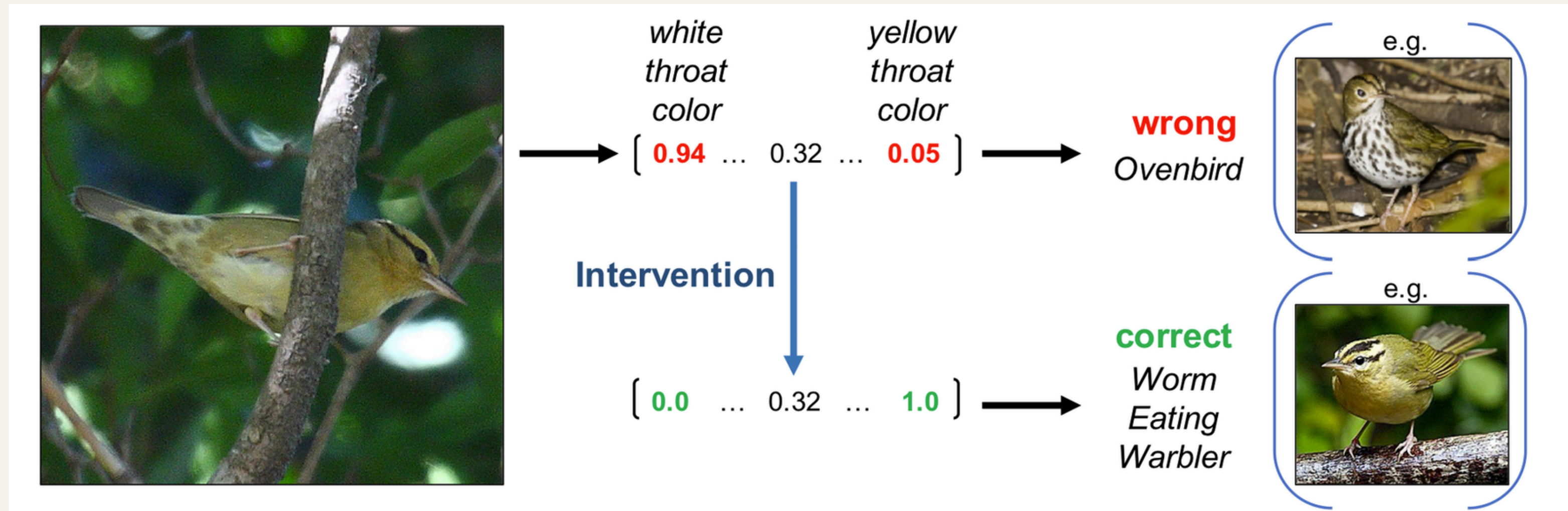
Why audio concept bottlenecks ?

Explaining models' predictions with interpretable concepts



Concepts are actionable, not just descriptive

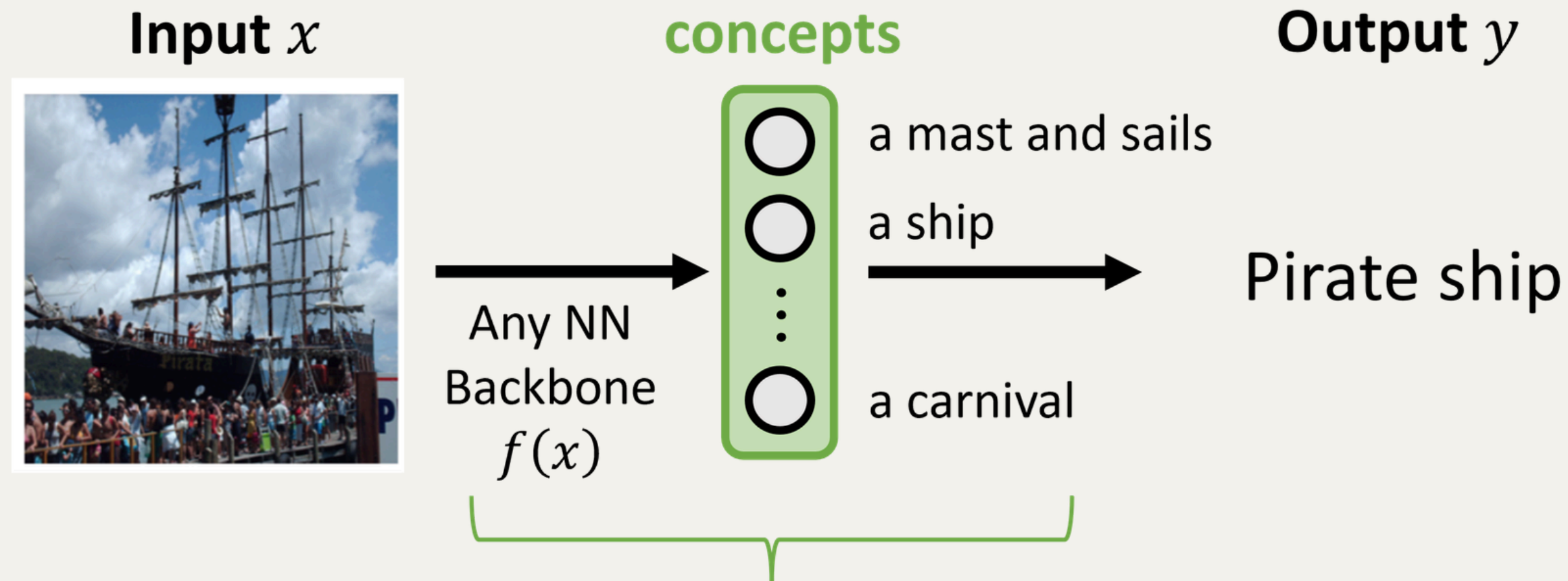
We can intervene directly on the prediction



- The major difficulty is that training them normally requires concept annotations for every sample

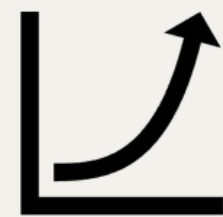
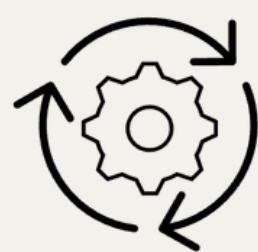
Label-Free CBMs

Removes the annotation requirement



Label-free CBM:

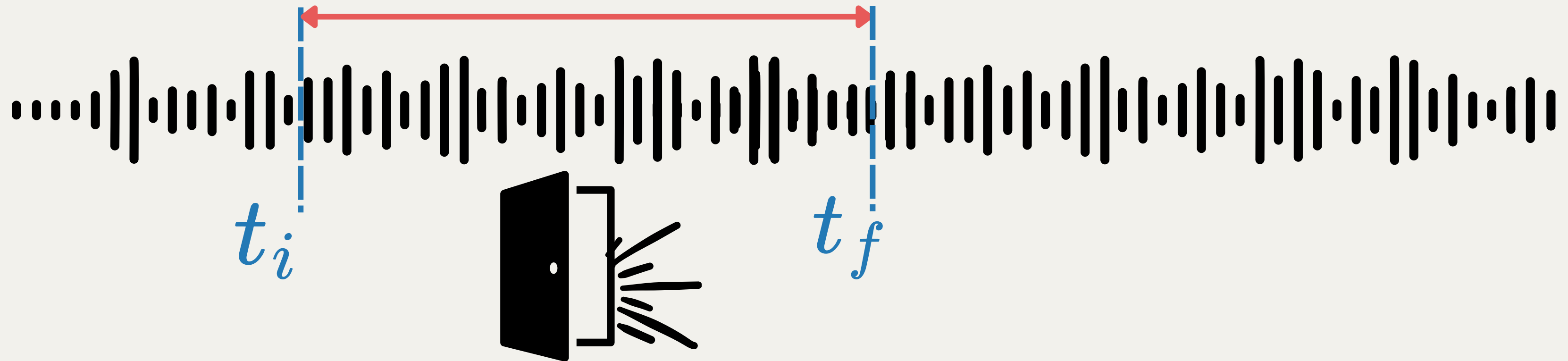
automated, scalable, efficient, no concept labels required



Why audio is different

1. Audio is temporal, images are static:

- A single global explanation can hide when the evidence appears



This motivates extending explanations temporally.

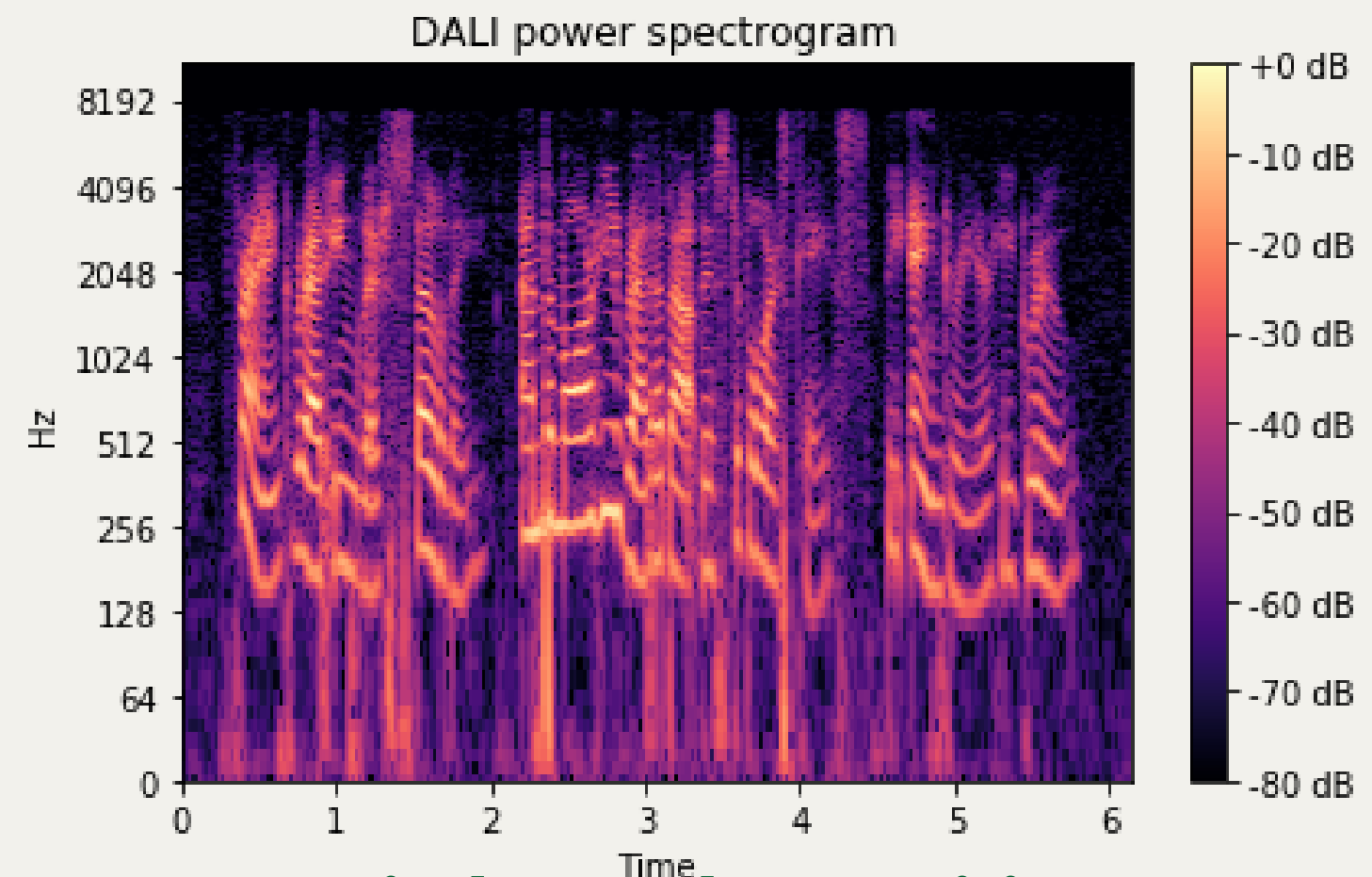
Why audio is different

2. Concept usefulness differs across modalities:

- In audio, broad perceptual terms can be highly informative
- Concepts such as high-pitched, rough, repetitive, impulsive, reverberant directly describe acoustic structure



Bright, High Contrast, Blurry...



Low pitch, Loud, Repetitive

This motivates an audio-specific concept vocabulary.

α -CBM Audio Label Free

Concepts generation

Original LF prompts

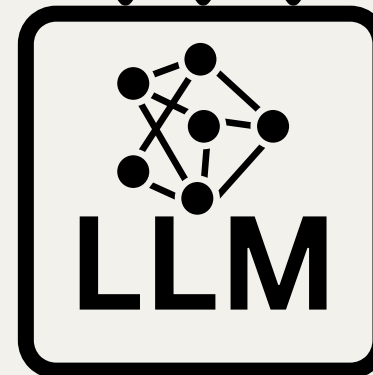
- “Most important features for recognizing a class”
- “Sounds commonly heard around a class”
- “Superclasses of a class”

Broad concepts (ours)

- Dataset-independent auditory attributes
- Pitch, timbre, roughness, attack, repetition
- No class names or event labels

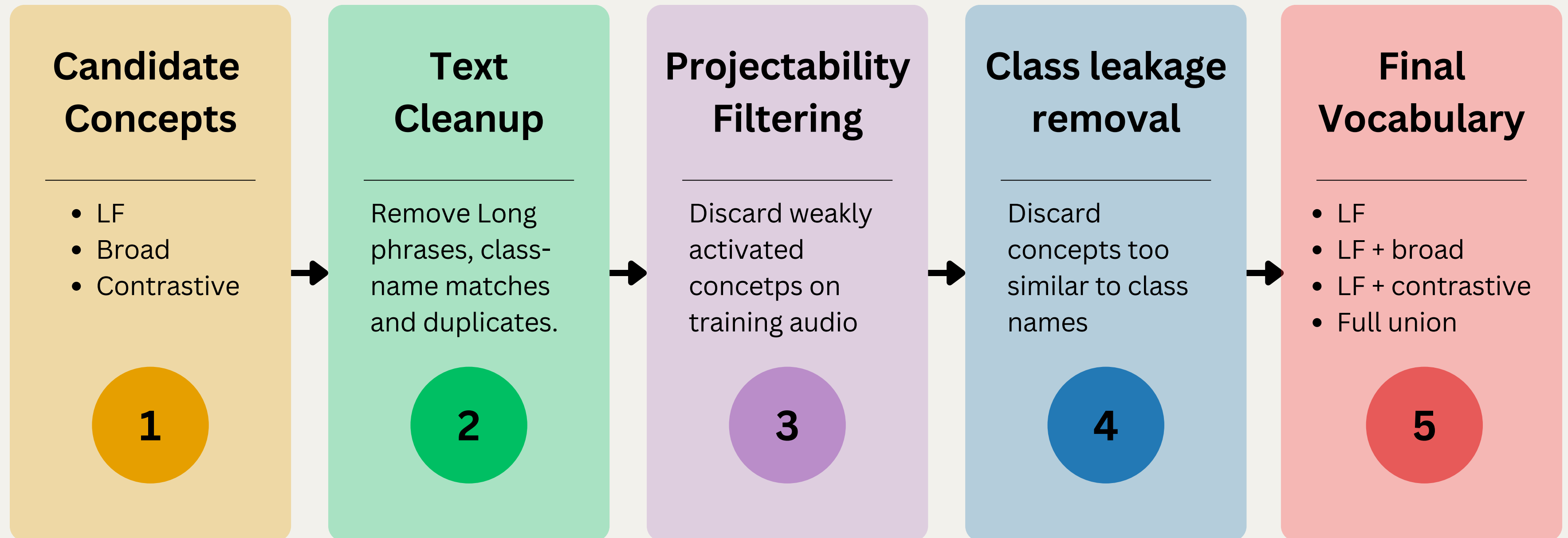
Contrastive concepts (ours)

- Generated for acoustically confusable class groups
- Class-name-free separating dimensions
- Targets discriminative acoustic differences



$$C_0 = C_{\text{LF}} \cup C_{\text{broad}} \cup C_{\text{contrast}}$$

Concept filtering



$$S_{\text{class}}(c) = \max_{y \in \mathcal{Y}} \text{sim}(c, y)$$
$$S_{\text{class}}(c) \geq 0.85 \Rightarrow \text{discard } c$$

Architecture

Label-free CBM For Audio

Step 1: Generate and filter **concept set**

Qwen3.5
Filtering

Target concepts

Distant cracking
Gusting howl
fabric rustle
engine rumble
applause
Wind gusts
Light purring

Image
Enc E_I
Text
Enc E_T
CLAP

Inner
product

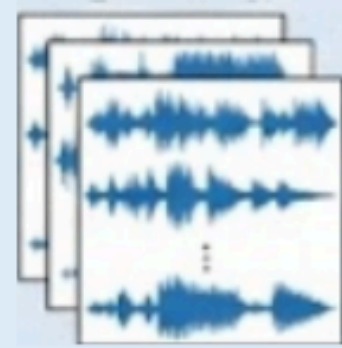
Concept Matrix

$E_I(x_1) \cdot E_T(t_1)$	...	$E_I(x_1) \cdot E_T(t_M)$
$E_I(x_2) \cdot E_T(t_1)$	...	$E_I(x_2) \cdot E_T(t_M)$
...	...	...
$E_I(x_N) \cdot E_T(t_1)$	...	$E_I(x_N) \cdot E_T(t_M)$

P

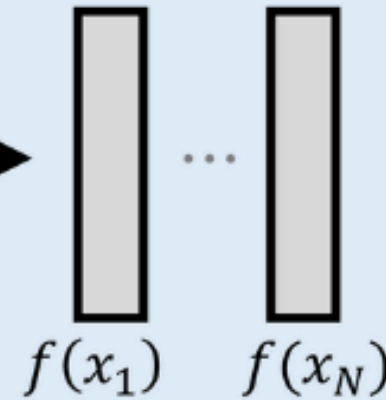
Step 2: Compute embedding $f(x)$ & **concept matrix P**

Input audio $x_1, \dots, x_n$



Backbone
 $f(x)$

Embeddings

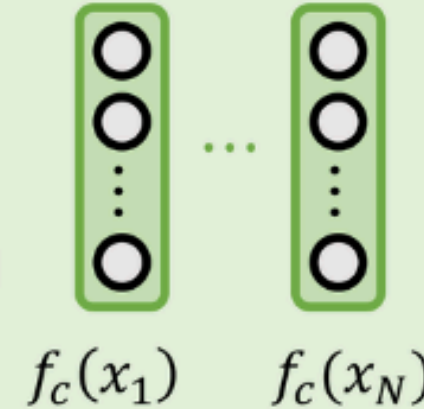


Step 3: Compute **CBL coefficient W_c** by max similarity between f_c and P

Concept bottleneck layer (CBL)

concept 1
concept 2
⋮
concept M

transform
 $f_c(x) = W_c f(x)$



Step 4: Train a sparse FC layer W_F on $f_c(x)$

Predicted class $\hat{y}$

$$\hat{y} = \operatorname{argmax}_{i \in \{1, 2, \dots, d_z\}} z_i$$

Bridge, car, ..., cloud

$\hat{y}_1 \quad \hat{y}_2 \quad \hat{y}_N$

FC layer
 $z = W_F f_c(x)$

(Notations)

$x_i \in \mathbb{R}^{d_0}$
 $f(x) \in \mathbb{R}^d$
 $W_c \in \mathbb{R}^{M \times d}$
 $f_c(x) \in \mathbb{R}^M$
 $z \in \mathbb{R}^{d_z}$
 $W_F \in \mathbb{R}^{d_z \times M}$

$$L(W_c) = \sum_{i=1}^M -\operatorname{sim}(t_i, q_i) := \sum_{i=1}^M -\frac{\bar{q}_i^3 \cdot \bar{P}_{:,i}^3}{\|\bar{q}_i^3\|_2 \|\bar{P}_{:,i}^3\|_2}.$$

Experimental Results

Broad audio concepts consistently improve vanilla LF-CBM

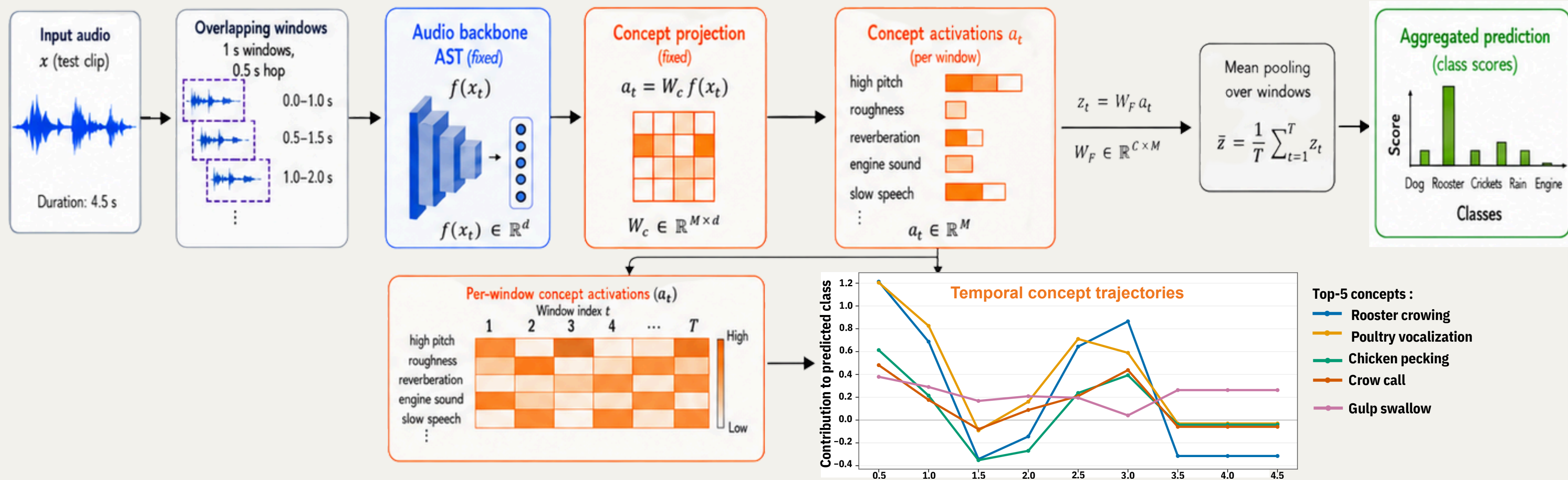
- **ESC-50** (*Piczak, 2015*) : environmental sound classification, 50 classes
- **UrbanSound8K** (*Salamon et al., 2014*) : urban sound classification, 10 classes
- **CREMA-D** (*Cao et al., 2014*) : speech emotion classification, 6 classes

Experimental Results

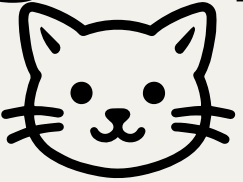
Dataset	Model	Acc.	F1	Kept/in.	Sp.
ESC-50	AST	93.50	93.28	—	—
	LF	93.00	92.97	596/603	91.78
	LF + broad	93.50	93.39	671/683	92.58
	LF + contrastive	93.25	93.01	689/700	92.82
	Full union	92.75	92.70	761/777	93.13
UrbanSound8K	AST	89.01	90.03	—	—
	LF	89.37	90.60	133/140	81.88
	LF + broad	89.84	90.94	212/224	86.04
	LF + contrastive	89.73	90.76	178/185	86.07
	Full union	89.84	90.78	255/269	87.69
CREMA-D (targeted)	AST	70.05	52.76	—	—
	LF	69.44	50.65	311/399	88.00
	LF + broad	69.78	50.69	481/598	91.34
	LF + contrastive	69.51	49.86	396/506	90.07
	Full union	70.25	51.58	548/701	92.79

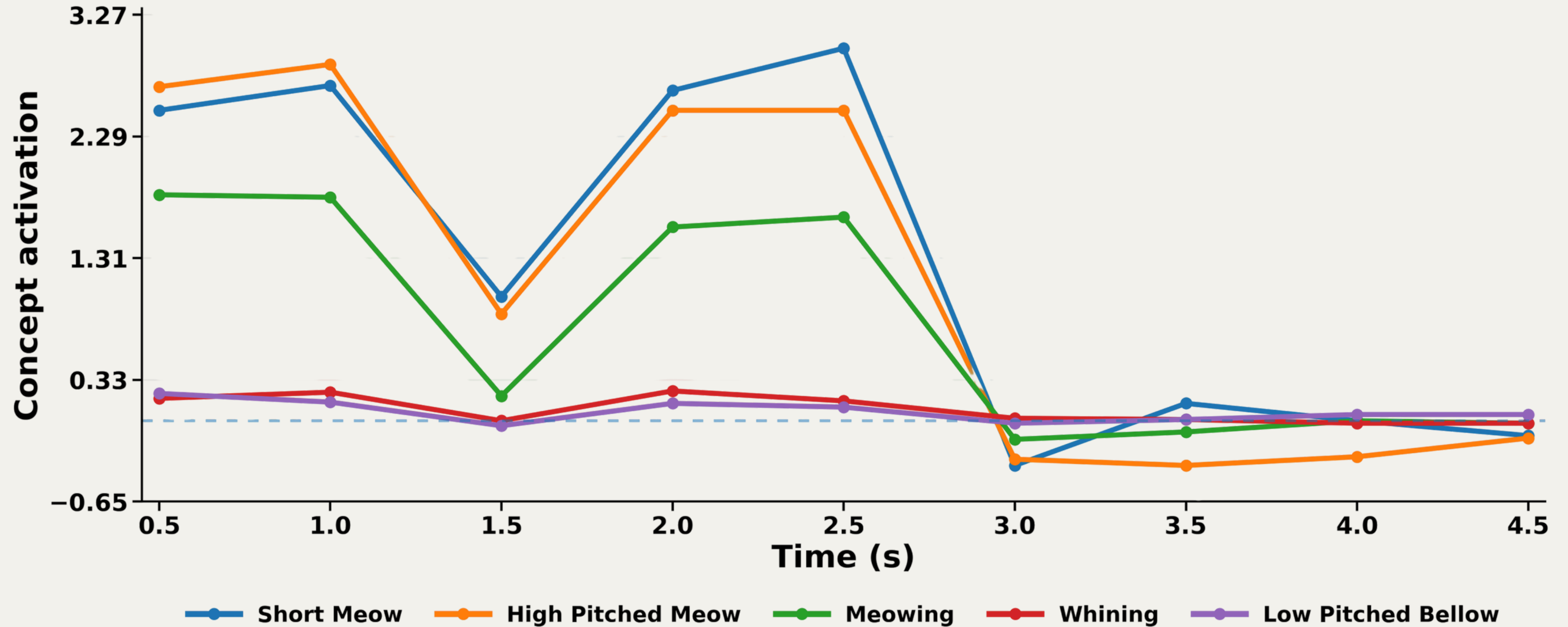
Extending Explanation temporally

Temporal explanations



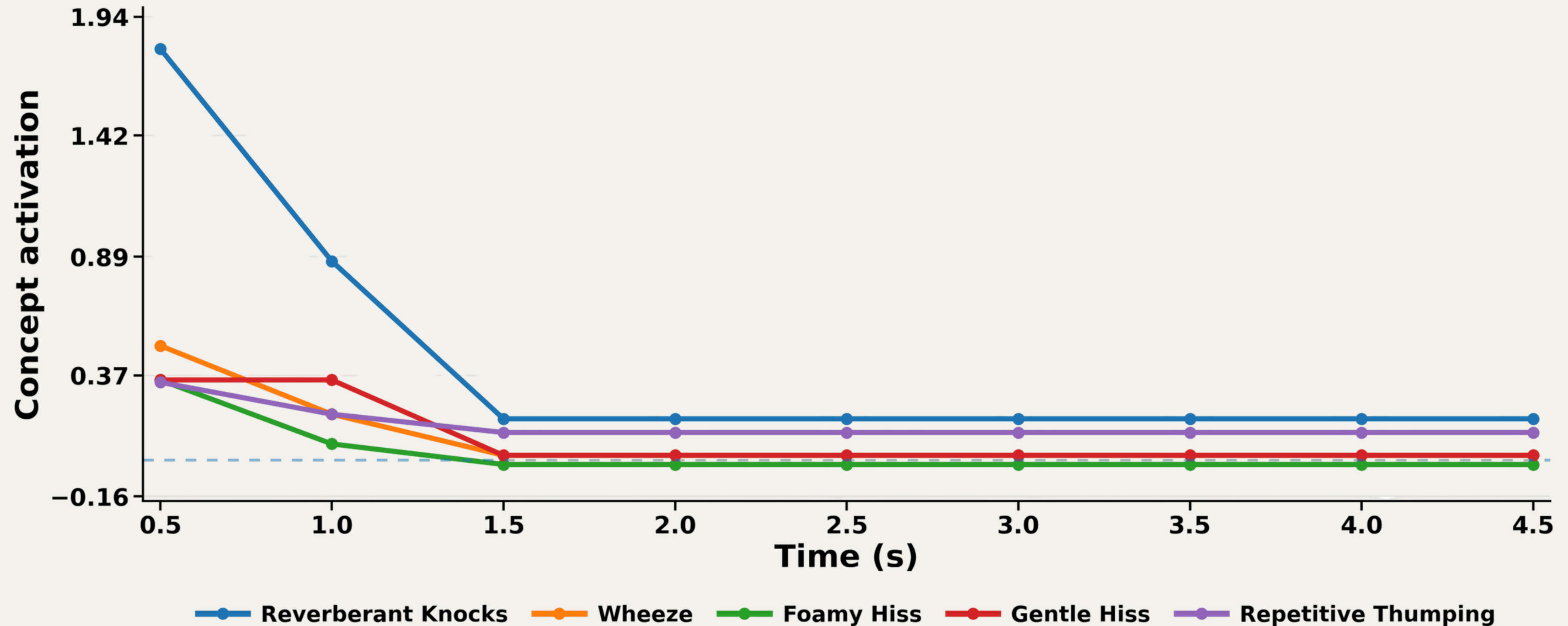
Extending Explanation temporally

Example : Cat 



Extending Explanation temporally

Example: Door knocking



Thank you

